PENETRATION TESTING REPORT

Footprinting & Network Scanning Phases

Week 2 | CYBERSECURITY | NETWORKWALKS

Field	Detail
Pentester Name (Cybersecurity Professional)	Prajapati Meet Ghanshyam
Program / Batch	B804 - Networkwalks
Date	12 November 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM2 (GHDB Footprinting) W2-PM5 (Zenmap Network Scanning)
Target(s)	networkwalks.com (client's own public website and DNS records) Tester's own LAN (10.128.13.0/24)
Permission / Authorization	Letter of Authorization for Cybersecurity Testing: **[add reference no. / status]**
Phases covered	Phase 1: Footprinting & Reconnaissance Phase 2: Network Scanning

1. Liability Disclaimer

These materials are for education and research purposes only. The practical work in this report was carried out to learn how attacks work in real-world systems so that systems can be defended better. Hacking is only legal when (a) you test a device or network that you own or a lab environment, (b) you have written and documented permission from the owner, or (c) you work as a security professional under a signed agreement with an agreed scope. Everything outside these cases is illegal. The tester accepts full responsibility for the actions described in this report, all of which were limited to passive and light information gathering within the scope below.

2. Introduction

This report covers the first two phases of a penetration test: footprinting/reconnaissance and network scanning. Three Week 2 project modules were completed: **W2-PM1** (footprinting networkwalks.com with six Kali Linux tools), **W2-PM2** (footprinting with Google Hacking Database dorks) and **W2-PM5** (host discovery with Zenmap). All outputs and screenshots shown were captured on 12 November 2026.

2.1 Scope and rules

In scope: the public website networkwalks.com and its DNS records, and the tester's own local network. Only passive and light tools were used. No exploitation, brute-force, login attempts or denial-of-service activity was performed. For PM2, search results were only viewed from Google and the public pages themselves; no credentials were entered and no camera controls were operated.

3. Tools Used

Tool	Purpose	Module
whois	Domain registration details, name servers	PM1
whatweb	Web technology fingerprinting	PM1
nslookup	DNS resolution of domain to IP	PM1
curl -I	HTTP response header inspection	PM1
wafw00f	Web Application Firewall detection	PM1
dnsrecon	DNS record enumeration	PM1
Google + Exploit-DB GHDB	Google dorks to find exposed devices and public files	PM2
Zenmap (Nmap 7.991)	Ping scan / host discovery and topology view	PM5
ipconfig (Windows)	Local IP address and subnet identification	PM5
Kali Linux (user: meet)	Platform used for PM1 tools	PM1

4. Activities Performed

4.1 Phase 1 - Footprinting & Reconnaissance (W2-PM1)

Target: **networkwalks.com**. Six built-in Kali Linux tools were run from the terminal. Each reveals a different part of the target's public footprint.

Task 1 - whois: domain registration details

```
$ whois networkwalks.com
```



Figure 1: whois output (lower part of the record)

Result: the name servers are NS6135.HOSTGATOR.COM and NS6136.HOSTGATOR.COM, so the domain is hosted with HostGator. The registrar abuse contact is abuse@godaddy.com (phone 480-624-2505). Domain status flags are clientDeleteProhibited, clientRenewProhibited, clientTransferProhibited and clientUpdateProhibited. DNSSEC is **unsigned**. The output also showed a notice that the WHOIS server is being retired in favour of RDAP.

Task 2 - whatweb: web technology fingerprint

```
$ whatweb networkwalks.com
```

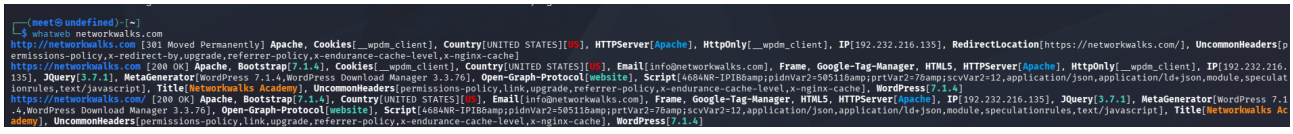


Figure 2: whatweb output

Result: the site first answers with a 301 redirect from HTTP to HTTPS, then HTTP 200. Detected: Apache web server, WordPress 7.1.4 with the WordPress Download Manager plugin 3.3.76, jQuery 3.7.1, Bootstrap, HTML5, Google Tag Manager, server IP **192.232.216.135** (country US), page title "Networkwalks Academy" and the public email address info@networkwalks.com.

Task 3 - nslookup: resolve the domain to an IP address

```
$ nslookup networkwalks.com
```



Figure 3: nslookup output

Result: networkwalks.com resolves to **192.232.216.135** (IPv4). The IPv6 answer 64:ff9b::c0e8:d887 is in the 64:ff9b::/96 range, which is a DNS64 translation of the same IPv4 address, so it is not a separate server. The query was answered by the local resolver 10.78.0.17.

Task 4 - curl -I: HTTP response headers

`$ curl -I https://networkwalks.com`



Figure 4: curl -I output

Result: HTTP/2 200. server: Apache; x-nginx-cache: WordPress (a caching layer in front of WordPress); a link header exposes the WordPress REST API at /wp-json/; a __wpdm_client cookie is set with the Secure and HttpOnly flags; referrer-policy is no-referrer-when-downgrade; the response date was Thu, 12 Nov 2026 03:54:32 GMT. HSTS, X-Frame-Options and Content-Security-Policy headers were not seen in this response.

Task 5 - wafw00f: Web Application Firewall detection

`$ wafw00f networkwalks.com`

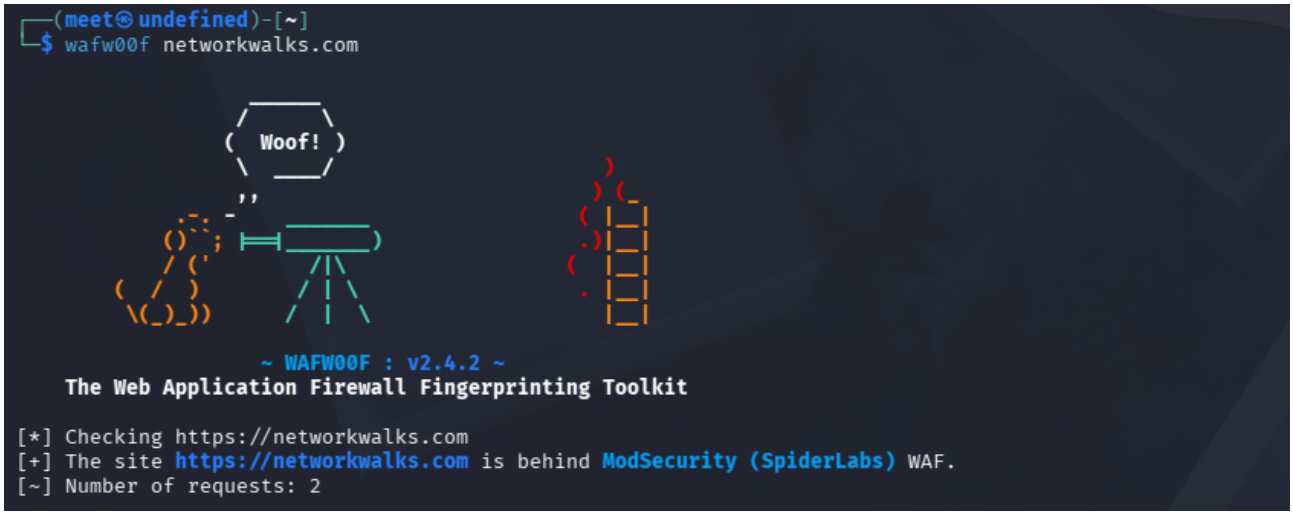


Figure 5: wafw00f output

Result: the site is behind a **ModSecurity (SpiderLabs)** WAF (wafw00f v2.4.2, 2 requests). Any later testing would have to account for this filtering.

Task 6 - dnsrecon: DNS enumeration

`$ dnsrecon -d networkwalks.com`

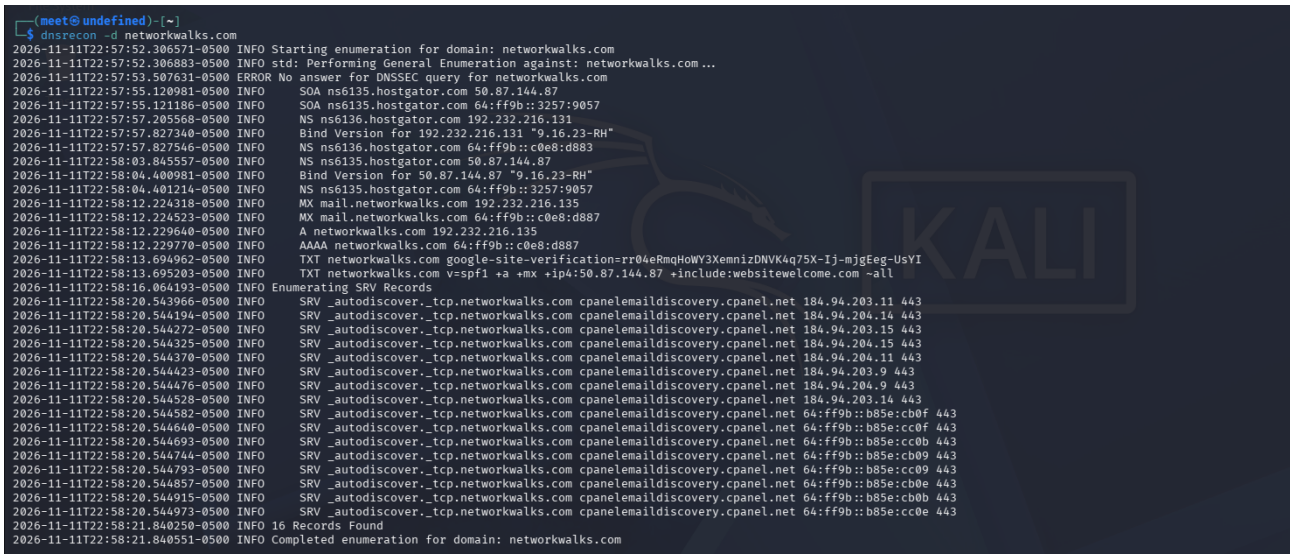


Figure 6: dnsrecon output

Record	Value found
SOA / NS	ns6135.hostgator.com (50.87.144.87), ns6136.hostgator.com (192.232.216.131)
DNS software	BIND version string "9.16.23-RH" returned by both name servers
MX	mail.networkwalks.com (192.232.216.135)
A / AAAA	192.232.216.135 / 64:ff9b::c0e8:d887
TXT (SPF)	v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all
TXT (verification)	Google site verification record present
SRV	_autodiscover._tcp pointing to cpanelemaildiscovery.cpanel.net (cPanel hosting)
DNSSEC	No answer to the DNSSEC query
Total	16 records found

Result: the DNS footprint shows HostGator name servers, cPanel based hosting and mail on the same IP as the website. The SPF policy ends in ~all (soft fail).

4.2 Phase 1 - Footprinting with GHDB (W2-PM2)

Google dorks from the Google Hacking Database (Exploit-DB) were used to find publicly exposed devices and files through Google only, without touching any target directly. Results were only viewed; no login was attempted and no camera controls were used. Identifying details of third-party cameras are not reproduced in this report.

Task 1 - Exposed security cameras

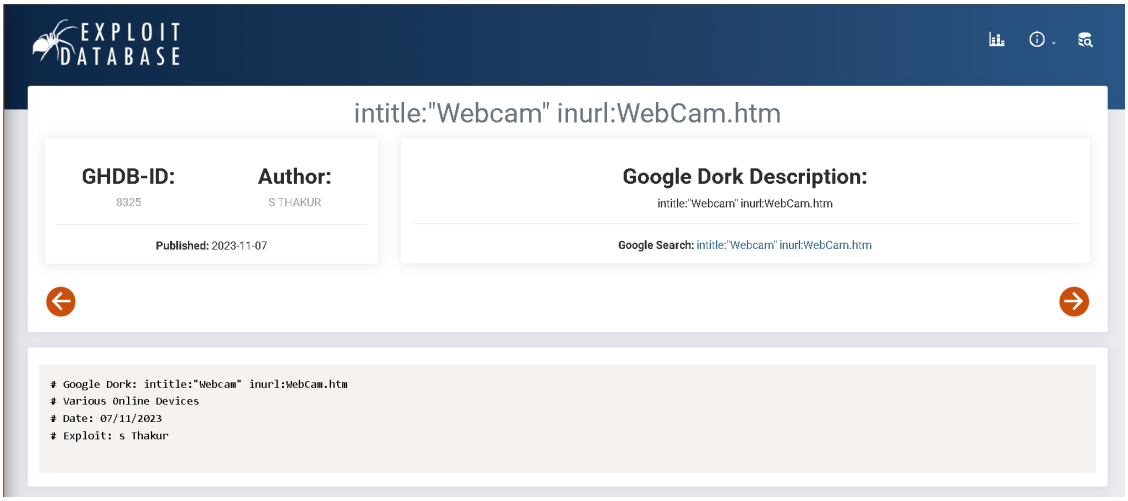


Figure 7: Exploit-DB GHDB entry (GHDB-ID 8325) for the dork used

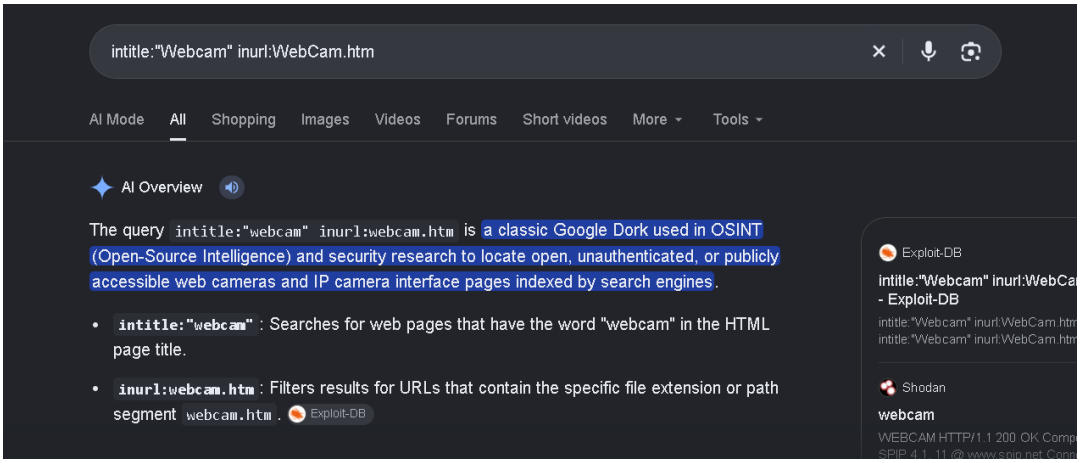


Figure 8: The dork searched on Google

Two exposed camera interfaces opened from the search results (screenshots trimmed to the product banner; both pages showed "Not logged in", i.e. no authentication was needed to view the stream):

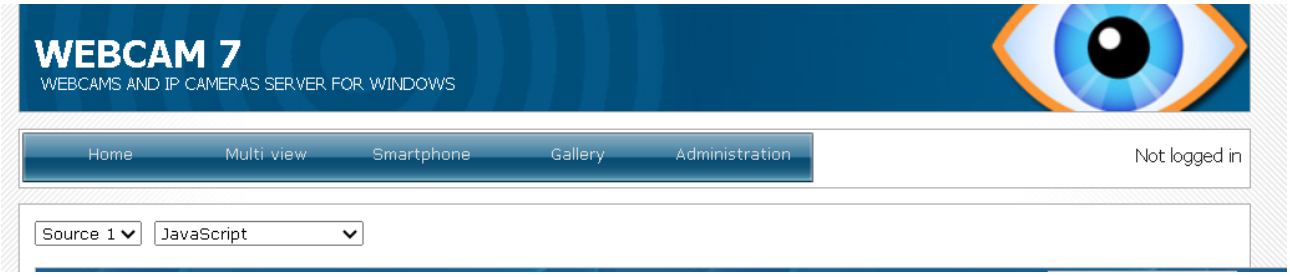


Figure 9: Exposed camera web interface (WEBCAM 7)

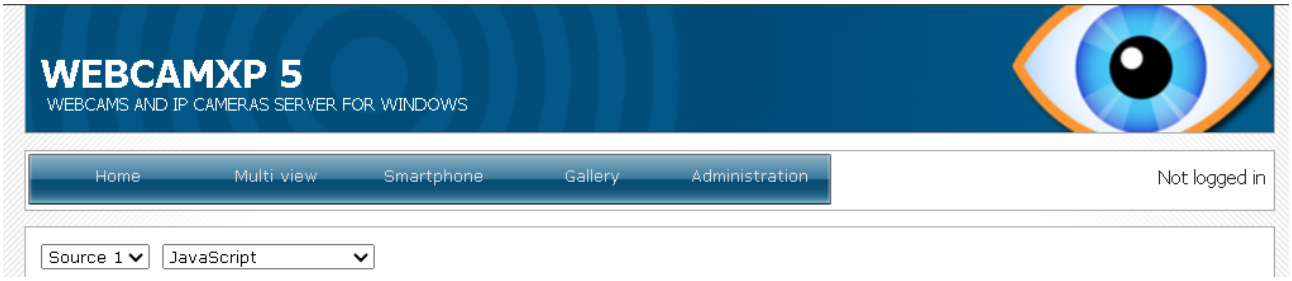


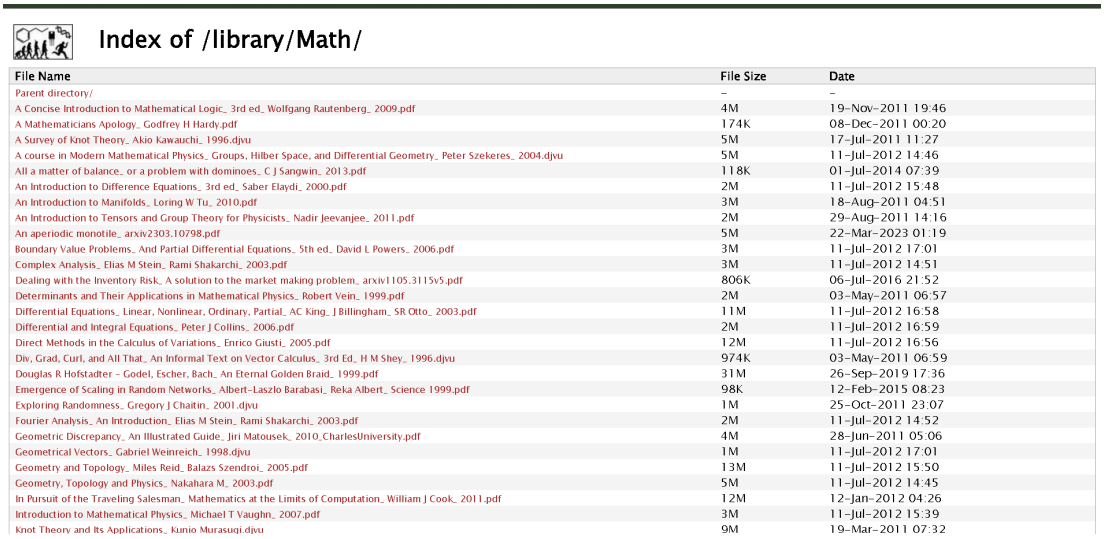
Figure 10: Exposed camera web interface (WEBCAMXP 5)

Table 1: Exposed camera links (target: 10)

No.	Link	Relevant Dork	Username / Password (if any)
1	[add URL - not visible in screenshot]	intitle:"Webcam" inurl:WebCam.htm	---
2	[add URL - not visible in screenshot]	intitle:"Webcam" inurl:WebCam.htm	---
3	[add link]	[add dork]	---
4	[add link]	[add dork]	---
5	[add link]	[add dork]	---
6	[add link]	[add dork]	---
7	[add link]	[add dork]	---
8	[add link]	[add dork]	---
9	[add link]	[add dork]	---
10	[add link]	[add dork]	---

Task 2 - Open directories with mathematics PDF ebooks

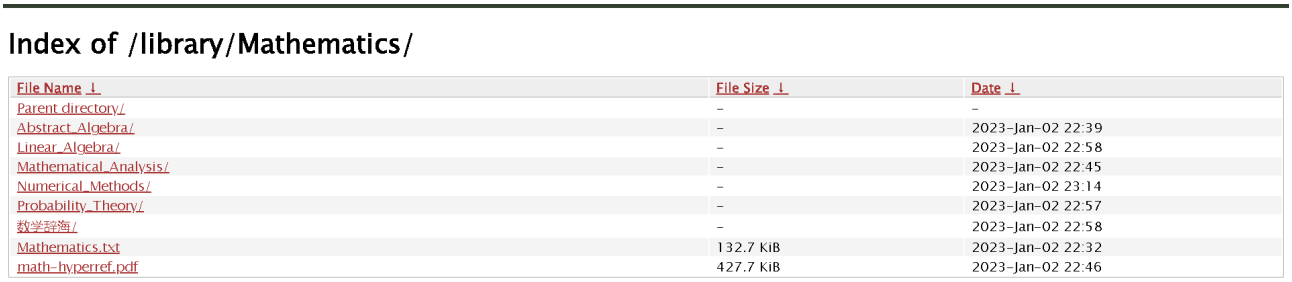
Directory-listing dorks (index of / parent directory + mathematics pdf) returned open directories hosting mathematics books in PDF format:



Index of /library/Math/

File Name	File Size	Date
Parent directory/	-	-
A Concise Introduction to Mathematical Logic_ 3rd ed_ Wolfgang Rautenberg_ 2009.pdf	4M	19-Nov-2011 19:46
A Mathematicians Apology_ Godfrey H Hardy.pdf	174K	08-Dec-2011 00:20
A Survey of Knot Theory_ Akio Kawauchi_ 1996.djvu	5M	17-Jul-2011 11:27
A course in Modern Mathematical Physics_ Groups, Hilber Space, and Differential Geometry_ Peter Szekeres_ 2004.djvu	5M	11-Jul-2012 14:46
All a matter of balance_ or a problem with dominoes_ C J Sangwin_ 2013.pdf	118K	01-Jul-2014 07:39
An Introduction to Difference Equations_ 3rd ed_ Saber Elaydi_ 2000.pdf	2M	11-Jul-2012 15:48
An Introduction to Manifolds_ Loring W Tu_ 2010.pdf	3M	18-Aug-2011 04:51
An Introduction to Tensors and Group Theory for Physicists_ Nadir Jeevanjee_ 2011.pdf	2M	29-Aug-2011 14:16
An aperiodic monotile_ arxiv2303.10798.pdf	5M	22-Mar-2023 01:19
Boundary Value Problems_ And Partial Differential Equations_ 5th ed_ David L Powers_ 2006.pdf	3M	11-Jul-2012 17:01
Complex Analysis_ Elias M Stein_ Rami Shakarchi_ 2003.pdf	3M	11-Jul-2012 14:51
Dealing with the Inventory Risk_ A solution to the market making problem_ arxiv1105.3115v5.pdf	806K	06-Jul-2016 21:52
Determinants and Their Applications in Mathematical Physics_ Robert Vein_ 1999.pdf	2M	03-May-2011 06:57
Differential Equations_ Linear, Nonlinear, Ordinary, Partial_ AC King_ J Billingham_ SR Otto_ 2003.pdf	11M	11-Jul-2012 16:58
Differential and Integral Equations_ Peter J Collins_ 2006.pdf	2M	11-Jul-2012 16:59
Direct Methods in the Calculus of Variations_ Enrico Giusti_ 2005.pdf	12M	11-Jul-2012 16:56
Div, Grad, Curl, and All That_ An Informal Text on Vector Calculus_ 3rd Ed_ H M Shey_ 1996.djvu	974K	03-May-2011 06:59
Douglas R Hofstadter - Godel, Escher, Bach_ An Eternal Golden Braid_ 1999.pdf	31M	26-Sep-2019 17:36
Emergence of Scaling in Random Networks_ Albert-Laszlo Barabasi_ Reka Albert_ Science 1999.pdf	98K	12-Feb-2015 08:23
Exploring Randomness_ Gregory J Chaitin_ 2001.djvu	1M	25-Oct-2011 23:07
Fourier Analysis_ An Introduction_ Elias M Stein_ Rami Shakarchi_ 2003.pdf	2M	11-Jul-2012 14:52
Geometric Discrepancy_ An Illustrated Guide_ Jiri Matousek_ 2010_CharlesUniversity.pdf	4M	28-Jun-2011 05:06
Geometrical Vectors_ Gabriel Weinreich_ 1998.djvu	1M	11-Jul-2012 17:01
Geometry and Topology_ Miles Reid_ Balazs Szendroi_ 2005.pdf	13M	11-Jul-2012 15:50
Geometry, Topology and Physics_ Nakahara M_ 2003.pdf	5M	11-Jul-2012 14:45
In Pursuit of the Traveling Salesman_ Mathematics at the Limits of Computation_ William J Cook_ 2011.pdf	12M	12-Jan-2012 04:26
Introduction to Mathematical Physics_ Michael T Vaughn_ 2007.pdf	3M	11-Jul-2012 15:39
Knot Theory and Its Applications_ Kunio Murasugi.djvu	9M	19-Mar-2011 07:32

Figure 11: Index of /library/Math/



Index of /library/Mathematics/

File Name ↓	File Size ↓	Date ↓
Parent directory/	-	-
Abstract_Algebra/	-	2023-Jan-02 22:39
Linear_Algebra/	-	2023-Jan-02 22:58
Mathematical_Analysis/	-	2023-Jan-02 22:45
Numerical_Methods/	-	2023-Jan-02 23:14
Probability_Theory/	-	2023-Jan-02 22:57
数学辞海/	-	2023-Jan-02 22:58
Mathematics.txt	132.7 KiB	2023-Jan-02 22:32
math-hyperref.pdf	427.7 KiB	2023-Jan-02 22:46

Figure 12: Index of /library/Mathematics/

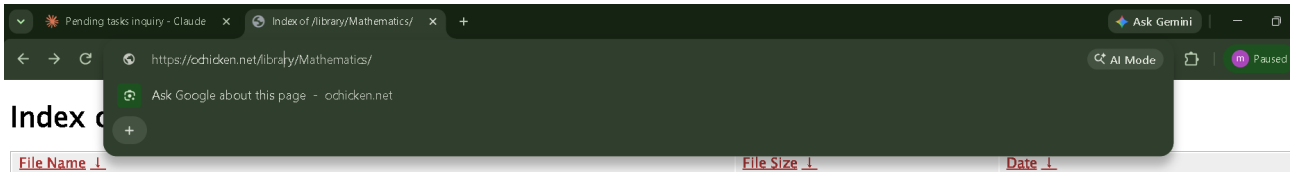


Figure 13: Address bar of the listing (host ochicken.net)

Table 2: Listings with downloadable mathematics ebooks (target: 10)

No.	Link	Relevant Dork	Username / Password (if any)
1	https://ochicken.net/library/Math/	[add dork]	---
2	https://ochicken.net/library/Mathematics/	[add dork]	---
3	[add link]	[add dork]	---
4	[add link]	[add dork]	---
5	[add link]	[add dork]	---
6	[add link]	[add dork]	---
7	[add link]	[add dork]	---
8	[add link]	[add dork]	---
9	[add link]	[add dork]	---
10	[add link]	[add dork]	---

4.3 Phase 2 - Network Scanning with Zenmap (W2-PM5)

Zenmap (the Nmap GUI) was used on the tester's own network to discover live hosts. This scan was limited to the tester's own LAN, as allowed by the authorization scope.

Task 1 - Install Zenmap

Zenmap was installed from the official Nmap website (https://nmap.org/download.html). The installed version is Nmap 7.991 as shown in the scan output below.

Task 2 - Local IP address and LAN subnet

`$ ipconfig`

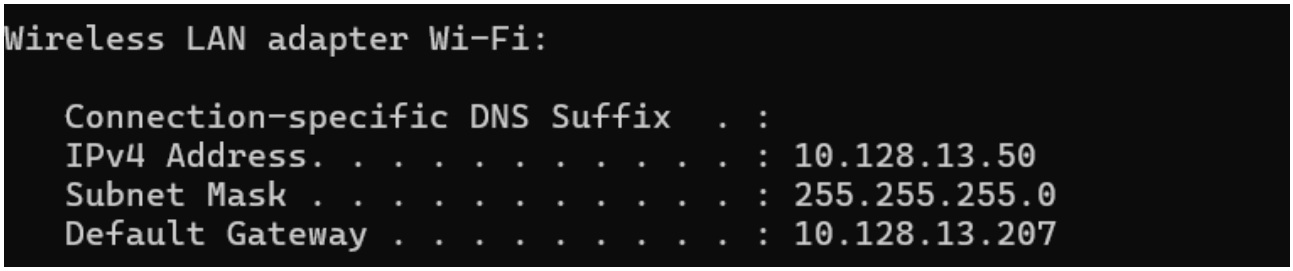


Figure 14: ipconfig output (Wireless LAN adapter Wi-Fi)

Item	Value
Local IPv4 address	10.128.13.50
Subnet mask	255.255.255.0 (/24)
LAN subnet	10.128.13.0/24
Default gateway	10.128.13.207

Task 3 - List of live hosts (Ping scan)

Target **10.128.13.0/24**, profile **Ping scan**, command **nmap -sn 10.128.13.0/24**.

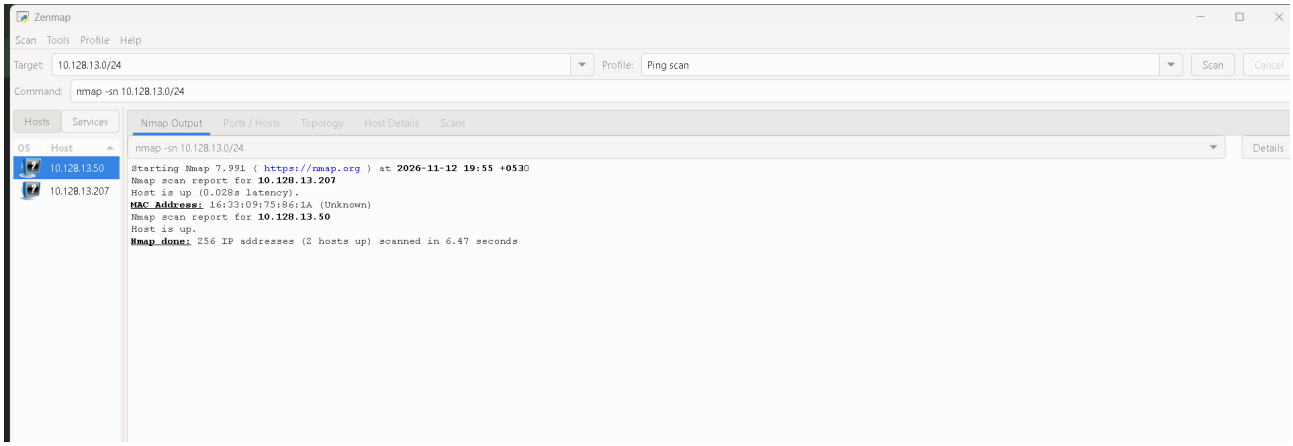


Figure 15: Zenmap ping scan result

Tasks 4, 5 and 6 - Number of live hosts, IP and MAC addresses

No.	IP address	MAC address	Note
1	10.128.13.50	[add - run ipconfig /all, Physical Address]	Tester's own PC (Nmap does not list the local MAC)
2	10.128.13.207	16:33:09:75:86:1A	Default gateway; vendor shown as Unknown; latency 0.028s

Task 4 answer: 2 hosts are live in the subnet (256 IP addresses scanned in 6.47 seconds), including the tester's own PC.
Task 5 answer: 10.128.13.50 and 10.128.13.207.

Task 6 answer: see table above.

Task 7 - Output topology

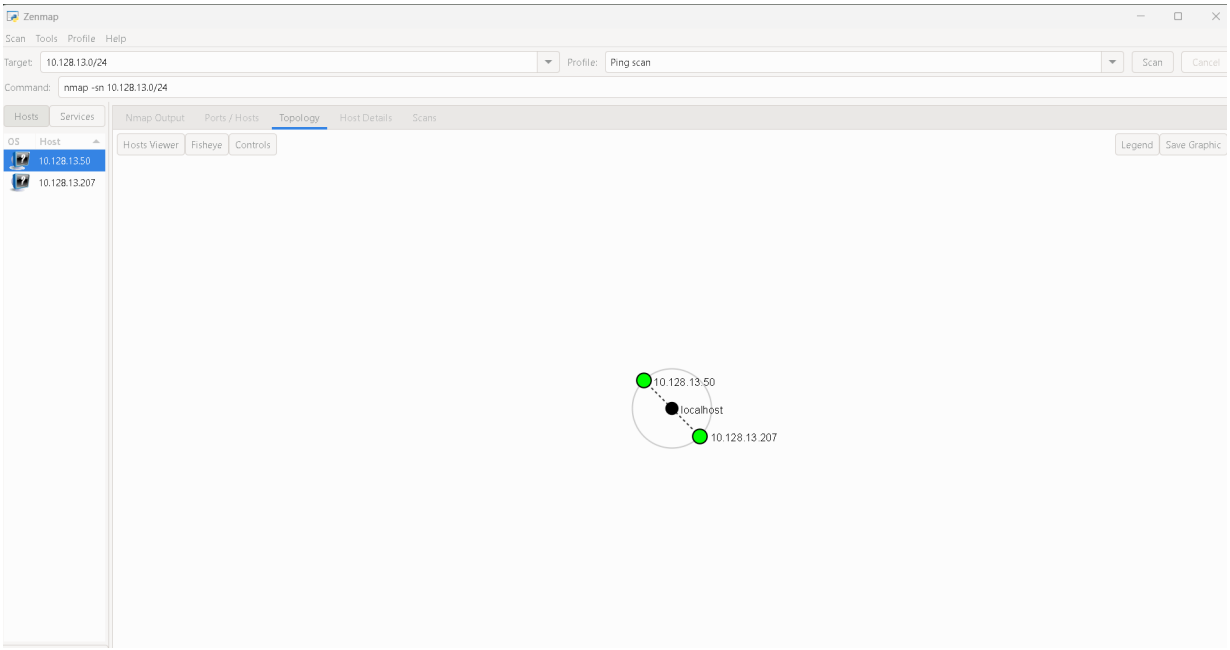


Figure 16: Zenmap Topology tab showing localhost with the two live hosts

The topology view shows the scanning PC (localhost) in the centre with both live hosts, 10.128.13.50 and 10.128.13.207, shown as green nodes, meaning fewer than 3 open ports were found (no port scan was run).

5. Findings and Risk Summary

The table below lists what an attacker could learn from the public information collected. These are information-exposure observations from passive recon, not confirmed vulnerabilities.

#	Observation	Source	Why it matters
1	WordPress 7.1.4 and Download Manager plugin 3.3.76 versions visible	whatweb	Exact versions can be matched against vulnerability databases.
2	Server IP 192.232.216.135 and hosting provider (HostGator, cPanel) revealed	whois, nslookup, dnsrecon	Shows the infrastructure and lets the server be scanned directly.
3	Public email address info@networkwalks.com exposed	whatweb	Can be used for phishing and spam.
4	WordPress REST API endpoint /wp-json/ advertised in headers	curl -I	Reveals an entry point that can be enumerated.
5	BIND version string 9.16.23-RH disclosed by name servers	dnsrecon	Version disclosure helps targeted attacks on DNS software.
6	DNSSEC not signed; SPF uses soft fail (~all)	whois, dnsrecon	Weaker protection against DNS spoofing and email spoofing.
7	Common security headers (HSTS, X-Frame-Options, CSP) not seen	curl -I	Fewer browser-side protections.
8	ModSecurity WAF present	wafw00f	Positive control: filters malicious web requests.
9	Open camera web interfaces viewable without login	GHDB	Shows how dorks expose devices that owners did not mean to publish.
10	Open directory listings hosting files	GHDB	Directory listing exposes file structure; sensitive files could leak the same way.

6. Recommendations

- Keep WordPress, themes and plugins updated and hide version information (generator tags, readme files).

- Disable or restrict the WordPress REST API and user enumeration where it is not needed.

- Hide server banners and DNS version strings (for example, set BIND "version" to a generic value).

- Add security headers: Strict-Transport-Security, X-Frame-Options, X-Content-Type-Options and Content-Security-Policy.

- Enable DNSSEC where the DNS provider supports it and move SPF to a strict policy (-all) after checking all legitimate mail sources.

- Use a role-based mailbox or contact form instead of publishing a direct email address.

- Keep the WAF rules updated and monitor logs for scanning activity.

- Never expose cameras or admin panels directly to the internet: put them behind a VPN, change default settings and require strong authentication. Disable directory listing on web servers.

- Defenders should run the same tools and dorks on their own domains regularly to see what an attacker would see.

7. Conclusion

Passive footprinting of networkwalks.com using six Kali tools produced a detailed profile of the target: registrar and name servers, hosting provider, server IP, web stack and versions, DNS and mail configuration, and the presence of a WAF. GHDB dorks showed how easily exposed devices and open directories can be found through Google alone. The Zenmap ping scan of the tester's own LAN (10.128.13.0/24) found 2 live hosts. The less information an organization leaks publicly, the harder every later stage of an attack becomes.